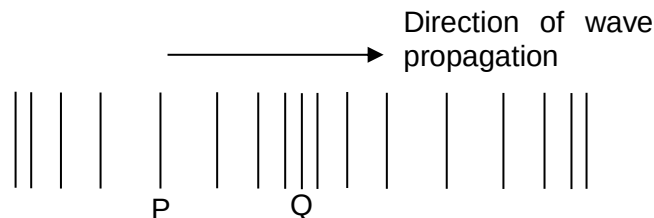


- 17 The diagram below shows a sound wave traveling from left to the right in air at a particular instant. Air particles P and Q are the centres of a rarefaction and a compression respectively. Which of the following describes the directions of motion of P and Q at this instant?

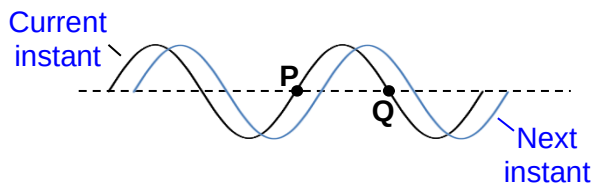


|          | Particle P   | Particle Q   |
|----------|--------------|--------------|
| <b>A</b> | to the right | to the left  |
| <b>B</b> | to the right | at rest      |
| <b>C</b> | to the left  | to the right |
| <b>D</b> | to the left  | at rest      |

Ans: C

Take direction of displacement to the right to be positive.

Since P and Q are centres of rarefaction and compression respectively, they both have zero displacement at this instant.



- 18 A vertical tube is completely filled with water. A small sound source of constant frequency is